

Generating Safe Recipes for Diabetes Management: An Empirical Study of Data Filtering and Model Scale

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Abstract

The reliability of Large Language Models (LLMs) remains a central bottleneck for safety-critical domains such as medically compliant recipe generation for diabetes. We frame this task as a constrained conditional generation problem and present a systematic study of how data curation and training scale affect safety outcomes, measured through Lexical Constraint Respect (LCR), a binary dietary compliance metric. Using FLAN-T5-base fine-tuned with LoRA, we find that data quality overwhelmingly dominates scale: a filtered 5K training set improves LCR by 24.9 percentage points over an unfiltered set of identical size. In contrast, reinforcement learning with binary rewards collapses performance, reducing our best supervised model from 84.2% to 31.3% LCR. These results demonstrate that meticulous data curation, rather than brute-force scaling or naive RL, is the most effective lever for improving safety and trustworthiness in constrained medical text generation.

1 Introduction

Large language models (LLMs) excel at open-ended text generation, but their reliability in safety-critical domains remains limited [1]. One such domain is diabetes-safe recipe generation, where violations of dietary guidelines—such as including added sugars or refined carbohydrates—can produce medically unsafe outputs [2]. We study this task as a constrained conditional generation problem requiring strict lexical compliance.

Millions of individuals rely on medically appropriate meal planning to manage blood glucose levels, and dietary errors can have direct clinical consequences. While LLMs are increasingly used to provide nutrition and meal-planning advice, evaluations show that models frequently hallucinate ingredients, omit restrictions, or offer guidance misaligned with medical standards [1]. A reliable system for generating diabetes-friendly recipes would represent a meaningful step toward trustworthy, personalized dietary assistance.

Naive prompting of LLMs rarely yields consistent constraint adherence: models often ignore forbidden ingredients, misinterpret substitutions, or generate incomplete recipes. Prior recipe-generation systems focus on personalization, style transfer, or cross-modal embeddings [3, 4], but none target strict medical safety constraints. Furthermore, typical training corpora like Food.com contain many unsafe recipes, making unconstrained fine-tuning counterproductive.

Existing health-aware LLM evaluations reveal poor alignment with clinical guidelines [1]. To our knowledge, no prior work has systematically studied the combined effects of data quality, dataset scale, and training strategy on safety outcomes in recipe generation. Reinforcement learning (RL) has

¹<https://www.overleaf.com/latex/templates/neurips-2019/tprktxwmqmgk>

been proposed for controllable generation, but reward sparsity and instability often lead to degraded outputs [5]. Thus, the field lacks principled evidence about which factors most effectively increase constraint compliance.

We hypothesize that data curation is a far stronger driver of safety than scaling dataset size or adding naive RL. To test this, we fine-tune FLAN-T5-base using LoRA [6] across multiple dataset sizes (1K–10K), filtered and unfiltered, and evaluate both supervised and RL-trained models. Our findings show that filtered datasets improve constraint satisfaction by 24.9 points over size-matched unfiltered data, while RL collapses performance. These results highlight the primacy of data quality in safety-constrained generation.

Our key contributions are:

- We present the first systematic study of data quality, dataset size, and RL training on diabetes-safe recipe generation.
- We construct a clinically motivated lexical constraint framework and filtered training corpus based on ADA guidelines.
- We evaluate seven supervised and RL models, demonstrating the dominant effect of data filtering on safety.
- We show that binary-reward RL is insufficient and often harmful for medically constrained text generation.

2 Motivating Choices

Our design choices reflect the clinical demands of diabetes-safe recipe generation and the computational limits of the course environment.

Safety as the primary objective. Because dietary violations have medical consequences, we operationalize safety through a deterministic Lexical Constraint Respect (LCR) metric built from ADA dietary guidelines [2]. This enables transparent dataset filtering and direct optimization.

Prioritizing data filtering over prompting. Prompt engineering alone does not reliably prevent unsafe ingredient hallucinations [1]. Early prompting trials confirmed frequent violations, motivating dataset-level supervision as a more stable and principled approach.

Parameter-efficient fine-tuning. We chose LoRA [6] due to limited GPU resources. LoRA rank $r = 8$ follows common PEFT best practices [7] and provided the best trade-off between stability and compute cost in preliminary trials.

Hyperparameter choices. We use a learning rate of 2×10^{-4} , batch size 8, and three epochs across all supervised experiments. These values follow prior T5 fine-tuning studies [8] and produced stable convergence without overfitting. Gradient accumulation enabled a larger effective batch size under memory constraints. RL experiments used REINFORCE [9] with a small reward coefficient to mitigate reward saturation.

Why explore RL? Although RL is widely used for controllable generation, its effectiveness for hard lexical constraints is unclear. Testing RL allowed us to evaluate whether binary safety rewards could improve constraint adherence beyond supervised learning. Our findings show they do not, offering a useful negative result for future work.

3 Related Work

Recipe generation. Neural recipe-generation research has explored cross-modal embeddings [3], user-personalized models [4], and comparative analyses of transformer-based generators [8]. These works focus on stylistic coherence or personalization rather than enforcing dietary safety constraints.

LLMs for nutrition and medical planning. Recent evaluations show that LLMs produce unsafe or clinically incorrect dietary recommendations, often ignoring medically relevant constraints [1]. Our work directly addresses these limitations by designing explicit constraint frameworks and quantifying the role of data curation.

Parameter-efficient fine-tuning. LoRA [6] and other parameter-efficient fine-tuning (PEFT) methods [7] offer substantial compute savings and have been shown to be effective for controlled or domain-specific text generation. We adopt LoRA to make systematic comparisons across dataset sizes feasible on a single GPU while maintaining competitive generation quality.

Reinforcement learning for controllable generation. Reinforcement learning has been applied to text generation tasks that optimize non-differentiable objectives such as factuality, style control, or safety constraints, typically using policy-gradient methods such as REINFORCE [9]. More recent work has shown that naive or sparse reward functions can lead to unstable training dynamics and degraded language quality, motivating the use of richer reward shaping or auxiliary critiques [5]. Our negative RL results are consistent with this prior work and highlight the difficulty of applying reinforcement learning to strict lexical or medical safety constraints in recipe generation.

Datasets. We draw from the Food.com recipes dataset [4], the Epicurious Food Ingredients and Recipes Dataset [10], and standard evaluation corpora such as RecipeNLG [11] to examine cross-dataset generalization. These datasets have been widely used in prior recipe-generation and food-domain NLP research, but they have not previously been filtered for medical safety or evaluated in constrained clinical generation settings such as diabetes-focused recipe adaptation.

4 Approach

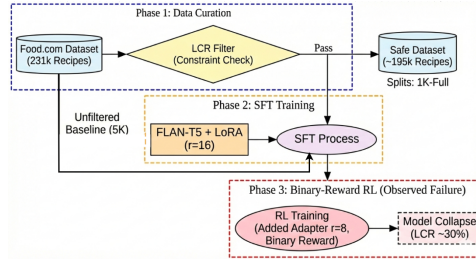


Figure 1: Overview of the three-phase pipeline: data curation, supervised fine-tuning, and binary-reward RL (observed failure).

Our approach investigates two key factors for safe recipe generation: data quality through filtering and model scale through varying training set sizes.

Problem Formulation. We frame diabetes-friendly recipe generation as a constrained conditional generation task. Given a dish name x (e.g., “chocolate cake”), the model must generate a recipe y containing ingredients and instructions that satisfy hard lexical constraints: avoiding harmful ingredients while including beneficial alternatives.

Lexical Constraint System (Our Contribution). We develop a keyword-based constraint system consisting of two manually curated lists derived from diabetes dietary guidelines[2]: $\mathcal{F}$ (267 forbidden keywords such as “white flour”, “sugar”, “butter”) and $\mathcal{R}$ (184 required keywords such as “whole wheat”, “olive oil”, “vegetables”). We define Lexical Constraint Respect (LCR) as:

$$\text{LCR}(y) = \left(\bigwedge_{f \in \mathcal{F}} \neg \text{Match}(y, f) \right) \wedge \left(\bigvee_{r \in \mathcal{R}} \text{Match}(y, r) \right) \quad (1)$$

where $\text{Match}(y, k)$ checks if keyword k appears as a complete word or phrase in recipe y (avoiding partial matches like finding “sugar” in “sugarcane”). A recipe passes if it contains zero forbidden keywords AND at least one required keyword. Using this LCR criterion, we filter the Food.com

dataset [4] to create safety-compliant training data. *All filtering code and keyword lists are our original contribution.*

Fine-Tuning with LoRA. We fine-tune FLAN-T5-base [8], a 248M parameter encoder-decoder model, using Low-Rank Adaptation (LoRA)[6]. LoRA adds trainable low-rank matrices to attention weights: $W = W_0 + \frac{\alpha}{r}BA$ where $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$. We use rank $r = 16$, scaling $\alpha = 32$, reducing trainable parameters to 0.36% of the full model. Implementation uses Hugging Face Transformers[12] and PEFT [7]. We train five model variants: **Safe LoRA 1K/5K/10K/Full** (filtered data with $N \in \{1000, 5000, 10000, 195000\}$) and **Unfiltered LoRA 5K** (no safety filtering).

Baseline Approaches. We establish two baselines using base FLAN-T5 without fine-tuning: **Baseline Vanilla** uses a simple prompt (“Generate a diabetes-friendly [dish] recipe”), while **Baseline Keywords** augments the prompt with explicit constraint examples (“Use: whole wheat, olive oil, vegetables. Avoid: white flour, sugar, butter”). These test whether prompting alone achieves constraint satisfaction.

Reinforcement Learning Experiments (Negative Result). To investigate whether RL could improve beyond supervised fine-tuning, we implemented REINFORCE [9] with binary LCR rewards, optimizing: $\mathcal{L}_{RL} = -\mathbb{E}_{\hat{y} \sim p_\theta} \left[\frac{r-b}{\sigma+\epsilon} \cdot \log p_\theta(\hat{y}|x) \right]$ where $r \in \{0, 1\}$ is the binary reward. Both experiments (Safe Full + RL, Unfiltered 5K + RL) resulted in catastrophic collapse—performance degraded from 84% to 28% LCR, with repetitive text generation. We hypothesize reward saturation caused this: when most samples pass, advantages approach zero, providing insufficient gradient signal. *This RL implementation is our contribution.*

Implementation. All code (training, filtering, evaluation) was implemented by us in PyTorch using Hugging Face libraries and is available in our submission.

5 Experiments

5.1 Data

Dataset. We use the Food.com Recipes and Interactions dataset [13], containing 231,637 recipes with dish names, ingredients lists, and step-by-step instructions. This serves as our training data source.

Task definition. Given a dish name as input, the model generates a complete recipe containing both ingredients and instructions. Input format: “Generate a diabetes-friendly [dish_name] recipe.” Expected output: “Ingredients: [list] Instructions: [steps].”

Example input/output:

Input: “Generate a diabetes-friendly grilled chicken recipe.”

Expected output: “Ingredients: boneless skinless chicken breast, olive oil, garlic cloves, lemon juice, whole wheat breadcrumbs, fresh herbs, black pepper. Instructions: 1. Marinate chicken in olive oil, garlic, and lemon juice for 30 minutes. 2. Coat with whole wheat breadcrumbs and herbs. 3. Grill over medium heat for 6-7 minutes per side until internal temperature reaches 165°F.”

Data filtering and splits. From 231,637 raw Food.com recipes, we apply LCR filtering to retain only recipes with zero forbidden keywords and at least one required keyword in their ingredients, yielding ~195,000 safe recipes. We create training sets of sizes 1K, 5K, 10K, and Full (~195K), each split 90/10 into train/validation. For comparison, we sample 5K unfiltered recipes without safety filtering.

Test sets. We evaluate on three test sets for cross-dataset generalization: (1) **RecipeNLG** (300 recipes sampled from the RecipeNLG corpus [11]), (2) **Epicurious** (300 recipes from the Epicurious dataset [10]), and (3) **Adversarial** (108 manually selected difficult dish names such as desserts and fried foods).

5.2 Evaluation Metrics

Lexical Constraint Respect (LCR). Our primary safety metric (defined in Section 5) returns 1 if a recipe contains zero forbidden keywords AND at least one required keyword; otherwise 0. We

report the percentage of recipes passing LCR. This binary metric reflects the hard constraint nature of dietary restrictions—a recipe either satisfies safety requirements or it does not.

ROUGE (Recall-Oriented Understudy for Gisting Evaluation). ROUGE [14] measures n-gram overlap between generated and reference recipes. ROUGE-1 counts unigram matches (individual word overlap), ROUGE-2 counts bigram matches (two consecutive words), and ROUGE-L finds the longest common subsequence. Higher scores indicate greater lexical similarity to human-written recipes. ROUGE scores range from 0 to 1, with typical values for generation tasks falling between 0.10-0.40 depending on task difficulty. Recipe generation is particularly challenging due to high variability in ingredient choices and instruction phrasing.

BERTScore. BERTScore [15] measures semantic similarity using contextual embeddings from RoBERTa-large. Unlike ROUGE’s exact word matching, BERTScore captures meaning: “chicken breast” and “boneless chicken” receive high similarity despite different words. We report F1 scores, which balance precision and recall. BERTScore typically ranges from 0.70-0.95, with scores above 0.80 indicating strong semantic alignment. Our baseline BERTScore of ~ 0.78 suggests reasonable semantic quality even for smaller models.

ROUGE and BERTScore are computed only for RecipeNLG and Epicurious (where references exist), not adversarial prompts.

5.3 Experimental Details

Supervised fine-tuning. All models use identical hyperparameters: AdamW optimizer with learning rate 3×10^{-4} , batch size 8, gradient accumulation of 4 steps (effective batch size 32), 3 epochs. We disable fp16 after preliminary experiments showed NaN gradients. LoRA configuration: rank $r = 16$, $\alpha = 32$, dropout 0.1, applied to query and value projections only.

Reinforcement learning setup. For RL experiments, we start from a supervised model (Safe Full or Unfiltered 5K) and apply a second LoRA adapter (rank $r = 8$, $\alpha = 16$, dropout 0.05) to avoid catastrophically overwriting learned patterns. We sample 2,500 prompts from the training set and train for 1 epoch with learning rate 1×10^{-5} (10 $\times$ lower than SFT). For each prompt, we generate a recipe using beam search (width 4), compute binary LCR reward ($r \in \{0, 1\}$), calculate standardized advantages using batch mean baseline, and optimize the REINFORCE objective. Generation during RL training uses the same beam search as inference to ensure reward signal matches evaluation conditions.

Generation. All models generate using beam search with width 4, max length 512 tokens, and early stopping.

Computational resources. All experiments run on a single NVIDIA T4 GPU (16GB memory).

5.4 Results

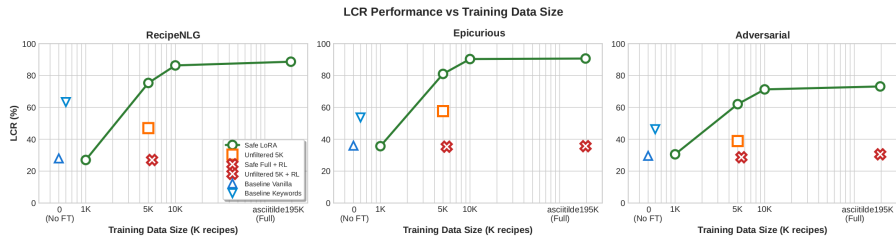


Figure 2: LCR performance across models and test sets. Safe models show clear scaling trends, while data filtering (Safe vs Unfiltered 5K) has dramatic impact. RL models collapse to baseline-level performance.

LCR results. Figure 2 visualizes LCR performance across all models. Table 1 provides exact values.

Key findings from LCR results:

Model	RecipeNLG	Epicurious	Adversarial	Average
Baseline Vanilla	28.0%	36.0%	29.6%	31.2%
Baseline Keywords	63.3%	53.7%	46.3%	54.4%
Safe LoRA 1K	27.0%	35.7%	30.6%	31.1%
Safe LoRA 5K	75.3%	81.0%	62.0%	72.8%
Unfiltered LoRA 5K	47.0%	57.7%	38.9%	47.9%
Safe LoRA 10K	86.3%	90.3%	71.3%	82.7%
Safe LoRA Full	88.7%	90.7%	73.1%	84.2%
Safe LoRA Full + RL	27.7%	35.7%	30.6%	31.3%
Unfiltered 5K + RL	27.0%	35.3%	28.7%	30.3%

Table 1: LCR performance across models and test sets.

(1) *Data filtering dramatically improves safety.* Safe LoRA 5K achieves 72.8% LCR versus 47.9% for Unfiltered LoRA 5K—a 24.9 point improvement with identical architecture and data size, demonstrating that data quality substantially impacts constraint satisfaction.

(2) *Diminishing returns with scale.* Scaling from 1K→5K yields +41.7 points (31.1%→72.8%), 5K→10K yields +9.9 points (72.8%→82.7%), but 10K→Full (19× more data) yields only +1.5 points (82.7%→84.2%).

(3) *Prompting is insufficient.* Baseline Vanilla (31.2%) and Baseline Keywords (54.4%) both fail to achieve reliable constraint satisfaction. The best fine-tuned model (Safe Full: 84.2%) outperforms Baseline Keywords by 29.8 points.

(4) *Cross-dataset generalization.* Safe Full achieves similar LCR on RecipeNLG (88.7%) and Epicurious (90.7%) despite Epicurious and RecipeNLG both being out-of-distribution, indicating learned constraints generalize across recipe sources.

(5) *Adversarial prompts are harder.* All models show 10-20 point lower LCR on adversarial prompts. Safe Full achieves 88-91% on standard sets but only 73.1% on adversarial, reflecting inherent difficulty of diabetes-friendly desserts and fried foods.

(6) *RL catastrophic failure.* Both RL models collapse: Safe Full + RL drops from 84.2% to 31.3% (-52.9 points). Both RL variants perform at baseline level, indicating complete model collapse rather than incremental degradation.

Content quality metrics. Figure 3 and Table 2 show ROUGE and BERTScore results.

Model	ROUGE-1	ROUGE-2	ROUGE-L	BERTScore F1
Safe LoRA 1K	0.106	0.016	0.077	0.784
Safe LoRA 5K	0.113	0.016	0.083	0.754
Unfiltered LoRA 5K	0.081	0.012	0.062	0.744
Safe LoRA 10K	0.156	0.024	0.111	0.771
Safe LoRA Full	0.196	0.033	0.126	0.785

Table 2: Quality metrics averaged over RecipeNLG and Epicurious. Scores in the 0.10-0.20 range for ROUGE and 0.74-0.79 for BERTScore are reasonable for challenging recipe generation.

Quality observations. Quality metrics improve substantially with scale: ROUGE-1 nearly doubles from 0.106 (1K) to 0.196 (Full), representing 84.9% relative improvement. However, metrics reveal different scaling behaviors. ROUGE-2 (bigram matching) remains very low across all models (0.012-0.033), indicating that consecutive word pairs rarely match exactly between generated and reference recipes—this reflects the high variability in how people phrase cooking instructions (e.g., "heat oil in pan" vs "warm olive oil in skillet"). ROUGE-L (longest common subsequence) shows more dramatic scaling: it remains poor for smaller models and unfiltered data (0.062-0.083 for 1K/5K/Unfiltered), but improves substantially for larger models (0.111 for 10K, 0.126 for Full), suggesting that longer phrase structures emerge only with sufficient training data. BERTScore shows less variation (0.744-0.785), indicating semantic alignment is relatively stable across models even when exact wording differs.

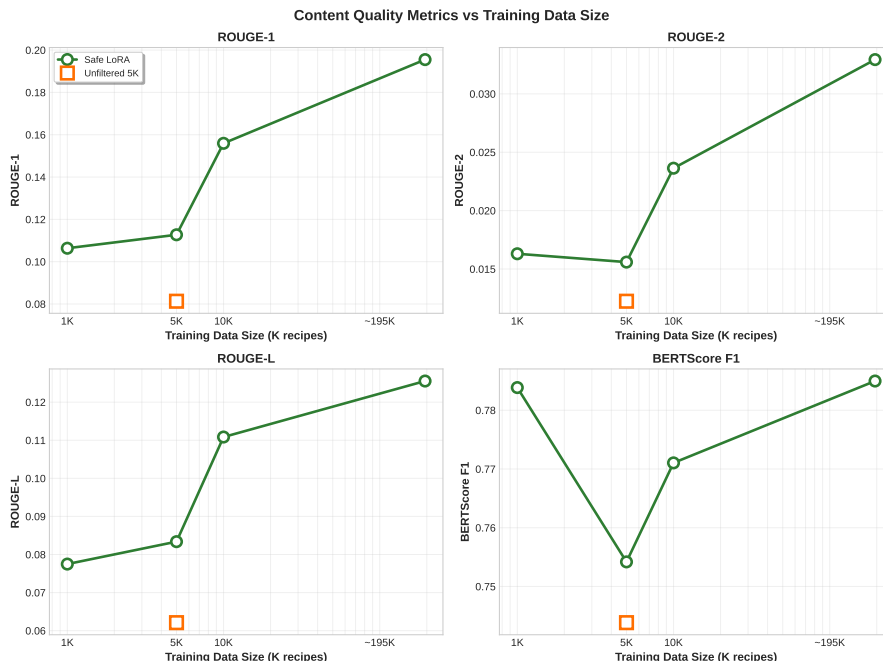


Figure 3: ROUGE and BERTScore across models (averaged over RecipeNLG and Epicurious). Quality improves with scale, and filtered data outperforms unfiltered despite identical size.

While these ROUGE scores appear low compared to tasks like summarization (which often achieve 0.30-0.50), recipe generation is inherently challenging due to high variability in ingredient choices and instruction phrasing—there are many valid ways to prepare the same dish. The consistency of BERTScore (staying around 0.75-0.79) despite varying ROUGE scores suggests our models maintain semantic quality even when exact wording varies.

Importantly, data filtering improves quality: Safe LoRA 5K outperforms Unfiltered LoRA 5K by +39.5% on ROUGE-1 (0.113 vs 0.081), +33.3% on ROUGE-2 (0.016 vs 0.012), +33.9% on ROUGE-L (0.083 vs 0.062), and +1.3% on BERTScore (0.754 vs 0.744). This indicates that filtering for safety does not sacrifice—and may even improve—generation quality, likely because filtered recipes represent higher-quality training examples with more appropriate ingredient substitutions.

6 Analysis

We examine generated recipe outputs to understand model behaviors, failure modes, and the impact of our design choices. We focus on representative examples that illustrate key patterns.

Baseline models produce incoherent or incomplete outputs. Without fine-tuning, models fail to generate proper recipes. Baseline Vanilla produces repetitive, structureless text: “*Grease and flour a 9x9x2-inch baking pan. Add the wet ingredients to the dry ingredients and mix well. Add the wet ingredients to the dry ingredients and mix well. Add the wet ingredients...*” continuing this loop with no actual ingredients listed. Even with explicit constraints in the prompt, Baseline Keywords often echoes the prompt verbatim (“*Use diabetes-friendly ingredients like: whole wheat flour, brown rice, olive oil...*”) rather than generating an actual recipe. These failures demonstrate that prompting alone—even with detailed constraint specifications—cannot achieve reliable recipe generation.

Small models lack structural competence. Safe LoRA 1K generates fragments rather than recipes. For example: “*Easy cheese danishes are a great option for diabetics. You can make these danishes as a side dish or as a side dish.*” This 18-word output lacks ingredients, instructions, measurements, or cooking steps—the model produces a description rather than a recipe. At 5K examples, structural competence emerges: all models consistently produce recipes with distinct “Ingredients:” and “Instructions:” sections, measurements, and step-by-step directions. However, these smaller models

still exhibit significant repetition: *“In a large bowl, combine onion, garlic, salt, pepper, garlic, salt, pepper, and pepper.”* The model understands recipe structure but struggles with content coherence.

Data filtering prevents unsafe ingredient choices. Comparing outputs from Safe vs. Unfiltered 5K models reveals filtering’s impact. For a caramel recipe, Unfiltered 5K generates: *“Ingredients: caramels, butter, sugar, salt, vanilla extract”* using multiple forbidden ingredients (butter, sugar) that would spike blood glucose. Safe 5K, trained on identical data size but filtered for safety, consistently avoids such violations and incorporates diabetes-friendly alternatives. This comparison isolates data quality effects—the difference in performance (72.8% vs 47.9% LCR) stems purely from training data curation, not model capacity.

Desserts reveal inherent task difficulty. Even our best model (Safe Full, 84.2% LCR) struggles with desserts. For a chocolate cake recipe, it generates: *“Ingredients: chocolate cake mix, butter, eggs, milk, vanilla extract, salt, pepper”* containing forbidden butter and lacking any diabetes-friendly substitutes like whole wheat flour or sugar alternatives. This pattern is consistent: desserts account for 33% of Safe Full failures. The challenge stems from fundamental constraint conflicts—traditional desserts require sugar and refined flour, and the model struggles to identify appropriate substitutions. In contrast, protein-based dishes like chicken or fish naturally accommodate diabetes-friendly ingredients (vegetables, olive oil), resulting in much lower failure rates across all models.

RL training causes catastrophic collapse. Both RL experiments failed dramatically. Safe Full + RL, starting from 84.2% LCR baseline, degraded to 27.7%—performing worse than untrained baselines. Generated outputs exhibit extreme degeneration. For example, when prompted for a hot chocolate recipe, the RL model produces only: *“Make a simple hot chocolate for one person.”* (9 words total). Other outputs show repetitive loops like *“Add chicken to the mixture and stir. Add chicken to the mixture and stir. Add chicken...”* repeating indefinitely.

We hypothesize this collapse stems from reward saturation: when the baseline model already achieves 84% pass rate, most training samples receive reward=1.0, causing advantages to approach zero and providing insufficient gradient signal for learning. Instead, the model converges to degenerate patterns that occasionally trigger keyword matches but lose all coherence.

This negative result demonstrates that binary pass/fail rewards are inadequate when baseline performance is already high. Future work should explore alternative reward designs, such as giving partial credit for avoiding forbidden ingredients even if required ingredients are missing, or training on progressively harder examples rather than the full distribution immediately.

7 Conclusion

We investigated diabetes-friendly recipe generation by fine-tuning FLAN-T5-base with LoRA, focusing on how data quality and scale affect constraint satisfaction.

Our key finding is that data quality dramatically outweighs quantity for safety-critical generation. Filtered 5K data achieves 72.8% LCR versus 47.9% for unfiltered 5K—a 24.9 point improvement with identical model size. We observe clear diminishing returns with scale: 1K→5K yields +41.7 points, but 10K→Full (19× more data) yields only +1.5 points. We identify a structural competence threshold at 5K examples and show that certain dish types (desserts) remain inherently difficult due to fundamental constraint conflicts.

Limitations. Although LCR provides a transparent and scalable measure of constraint satisfaction, it is inherently lexical and limited by the coverage of our ingredient keyword lists. This evaluation cannot assess cookability/taste (which would require human evaluation), portion sizes, or overall nutritional appropriateness, and may miss unsafe ingredients when they appear as rare synonyms, brand-name products, or implicit references (e.g., “cake mix”). Consequently, we avoid claiming full clinical safety and instead interpret our results as evidence about keyword-level adherence under both standard and adversarial prompting.

Future work. Semantic constraint checking using embeddings could improve beyond keyword matching. Repetition penalties could enhance fluency. Alternative RL reward designs (partial credit, shaped rewards) might avoid collapse. Nutritional analysis and structured human evaluation, particularly by domain experts, would provide a more complete safety assessment.

Our work demonstrates that for safety-critical generation, careful data curation yields better returns than simply scaling dataset size—an insight with broad implications for constrained generation in medical, legal, and other high-stakes domains.

8 Team Contributions

Marcelle developed and refined diabetes-specific lexical safety constraints and integrated them into the LCR evaluation and data-preparation pipeline. She also implemented data-prep/splitting utilities and executed key training and evaluation runs for the baseline and unfiltered-context models.

Ella implemented the FLAN-T5 baseline recipe-generation pipeline using the unfiltered Food.com data set, managing data loading, prompt formatting, evaluation scripts, and output processing. She trained the supervised models on the filtered data set and assisted in configuring the RL experiments on the filtered model.

Danielle conducted the full data ablation study and trained seven comparative models (full dataset, 1K, 5K, 10K, unfiltered 5K, and two RL variants), ensuring standardized hyperparameters across experiments. She also redesigned and expanded the evaluation framework (ROUGE, BERTScore, LCR, adversarial prompts) and created the project’s diagrams and figures.

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